Remote Sensing Semantic Change Detection With Multidimensional Spatiotemporal Scanning

Zhong Chen, Hanruo Chen, and Haodong Mu

***Abstract*—Remote sensing semantic change detection (RSSCD), which identifies pixel-level ground semantic changes by analyzing multitemporal data, is widely used in fields such as urban planning and disaster monitoring. However, most of the current datasets only contain two temporal phases, which provide insufficient information, leading to overlooking areas with subtle changes and mistakenly detecting nonchange areas with significant differences. Moreover, changes with diverse scales and directional distributions make it difficult to localize them accurately. Besides, class imbalance further degrades detection performance. To address these issues, this article proposes a novel Mamba-based multidimensional spatiotemporal scanning method. First, temporal interpolation is applied to bitemporal images, enabling high-dimensional spatiotemporal feature extraction through global interaction with linear complexity. Second, the shallow feature extraction module in the above model is improved, and a set of diagonal scanning directions is added to better capture multiscale (MS) and omnidirectional changes. Finally, a consistency dynamic weighted loss (CDW Loss) is introduced to mitigate class imbalance. The proposed method achieves significant improvements in the separated Kappa (SeK) metric, a comprehensive indicator for semantic change detection, outperforming the current state-of-the-art method (ChangeMamba) by 17.67%, 4.54%, and 20.80% on the Landsat–SCD, SECOND, and Hi-UCD datasets, respectively, in the absence of the proposed CDW Loss.**

***Index Terms*—Mamba, remote sensing, semantic change detection.**

I. Introduction

REMOTE sensing images are valuable data collected by sensors mounted on platforms, such as satellites, aircraft, or drones. Through analysis of visible remote sensing images, researchers have achieved significant tasks, such as object localization, instance segmentation, semantic segmentation, and change detection. With the advancement of computer technology and the overall improvement in computational capabilities, the performance of these tasks has been continuously optimized and gradually applied in various fields, including urban expansion assessment, forest cover statistics, desertification control, flood prevention, and agricultural development planning. Fundamentally, most of the above-mentioned tasks can be summarized as monitoring whether the land cover categories on the surface of important areas have changed, that is, remote sensing semantic change detection (RSSCD).

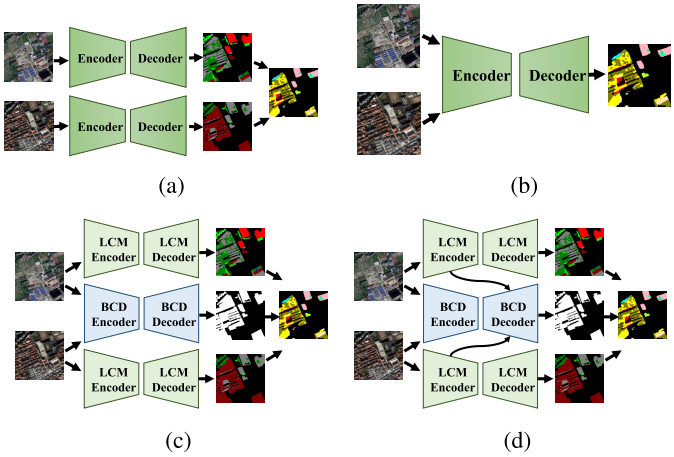


Fig. 1. Four common architectures of RSSCD. (a) Strategy 1. (b) Strategy 2. (c) Strategy 3. (d) Strategy 4.

Early research work on remote sensing change detection mainly focused on remote sensing binary change detection (RSBCD). As a simplified task of RSSCD, RSBCD is a pixel-level binary classification task used to identify whether semantic-level changes have occurred at each pixel location of the same place at different times. However, RSBCD cannot determine the land cover categories before and after the change, which is not conducive to further refining the change results. In general, what RSSCD needs to address is the issue of how to obtain multiple outputs through multiple inputs. Daudt et al. [1] proposed four feasible task strategies for RSSCD as shown in Fig. 1. These strategies encompass the basic structures of most current RSSCD models, but each of them has its own limitations.

The model based on the strategy of Fig. 1(a) indirectly obtains the semantic change results by performing semantic segmentation on multitemporal images, but the performance is the worst; this is because there are a large number of regions with unchanged semantics between the two temporal images. These regions can be used to enhance the segmentation ability of the model, but have not been utilized; that is,

Received 21 May 2025; revised 22 October 2025 and 1 March 2026; accepted 17 April 2026. Date of publication 22 April 2026; date of current version 19 May 2026. This work was supported in part by the Civil Aerospace Technology Pre-Research Project of China's 14th Five-Year Plan under Grant D040404 and in part by the Key Laboratory of Target Cognition and Application Technology under Project 2023-CXPT-LC-005. *(Corresponding author: Haodong Mu.)*

The authors are with the School of Artificial Intelligence and Automation, National Key Laboratory of Multispectral Information Intelligent Processing Technology, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: henpacked@hust.edu.cn; m202373675@hust.edu.cn; mhd2506346628@163.com).

Digital Object Identifier 10.1109/TGRS.2026.3686361

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

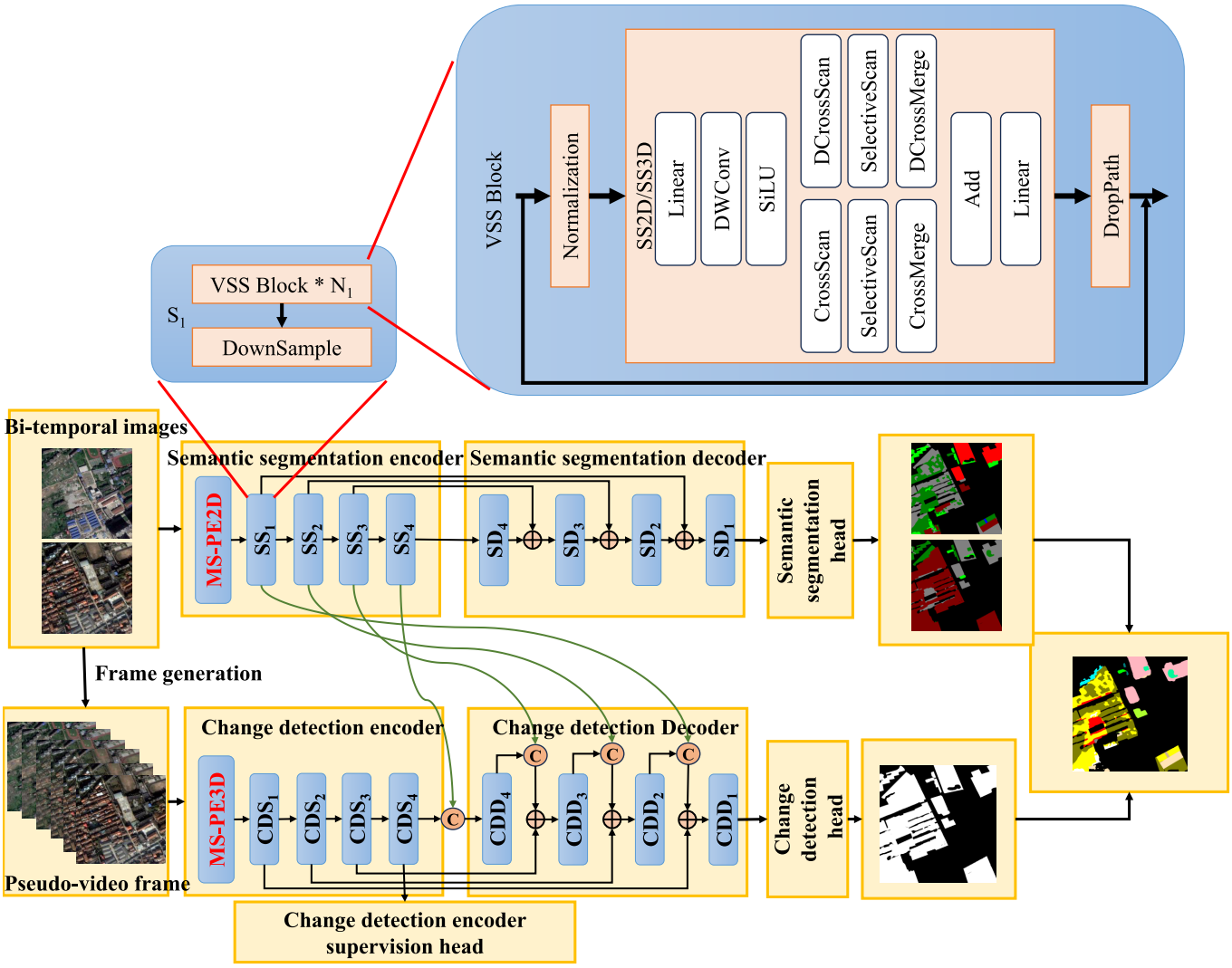


Fig. 2. Overall structure of our RSSCD model.

the inline information between the two temporal images has been ignored. The model based on the strategy of Fig. 1(b) directly combines the input multitemporal images into a single input. It does this by adding, subtracting, or splicing along the channel. Then, it obtains the semantic change result through semantic segmentation. However, if the test set has some changes that are not included in the training set, the model's test results are difficult to generalize to unknown change categories. The model based on the strategy of Fig. 1(c) adds a binary change detection branch to the model based on the strategy of Fig. 1(a) and masks the result of this branch onto the images from semantic segmentation. However, semantic inconsistencies may exist in the nonchanging regions. The direct mask ignores such inconsistencies, and at the same time, this operation also overlooks the correlation between the branch of remote sensing semantic segmentation (RSSS) and RSBCD. The model based on the strategy of Fig. 1(d) integrates the features extracted from the encoder of segmentation branch into the decoder of the binary change detection branch based on Fig. 1(c), thus associating the segmentation task with binary change detection. However, the problem of semantic inconsistency in the nonchanging regions still remains. We will build a model based on the fourth structure and solve the problem of inconsistent land cover types that occur in the unchanged regions determined by the RSBCD branch, based on the loss function.

In addition to the above strategies, RSSCD can also be regarded as a sequence task. Graves et al. [2] introduced the recurrent neural network (RNN) into the RSSCD to associate multiple remote sensing images with time series relationships and obtain the features of multitemporal images along the time dimension [3]. The multitemporal images in the dataset of RSSCD are inherently nonlinear due to various atmospheric scattering conditions, complex light scattering mechanisms, and variability of the same category at different times. The RNN for sequence tasks can also be used to process nonlinear continuous changes. This method first uses convolution to extract the local features of the multitemporal images. Then, taking these features as input, they are fed into the RNN according to the chronological order of image acquisition. After obtaining the output results, a semantic change detection map is obtained through a fully connected layer (FCL). How-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

ever, this method can only output direct results in detecting semantic changes. When it comes to the type of change that exists in the test set but not in the training set, its detection accuracy will be significantly reduced. In specific experiments, the researchers compared the accuracy of three RNN structures in RSSCD, including the fully connected neural network (FCN), the gated recurrent unit (GRU) [4], and long-short-term memory (LSTM) [5]. Finally, the experiment proved that LSTM is more effective.

However, most of the RSSCD datasets only provide bitemporal images, which can only offer the RGB color information and relative position information of various targets and regions. This is not conducive to the full extraction of temporal dimension information by methods based on RNNs. As a result, these models are unable to recognize changes with small differences and nonchanges with large differences. Therefore, as shown in Fig. 2, we first perform frame interpolation on the bitemporal input images of the remote sensing image semantic change detection dataset to obtain a sequence of frames. In other words, we enrich the input data with temporal dimension information. Then, we scan along multiple dimensions on this sequence of frames to extract more detailed semantic change features of the same location at different times, which helps the model improve its perception of changes and recognition of nonchanges.

In the latest research, Peng et al. [6] explored the issue of spatial correlation of the input image pairs and how to deal with the problem of class imbalance in the dataset. Regarding the former, they proposed a model based on the interaction and association between features of the same layer. To address the problem that the features of multitemporal images cannot be fully fused, they performed a subtraction operation on the features at each scale of the encoder and sent the results to the decoder at the same level. Meanwhile, to extract global features, they also applied self-attention to the feature connection of the encoder and decoder. Finally, semantic segmentation results are generated for the two images, respectively, to complete RSSCD. During training, to solve the class imbalance problem, these researchers combined the advantages of dice loss [7] and focal loss [8] to define a new class loss function.

However, the computation based on the Transformer has a time complexity of $O(n^2)$, which is not conducive to the training and inference of the model. Therefore, we use Mamba, which can also achieve global information interaction and has a linear computational complexity, as the main method for model feature learning. Regarding the problem of class imbalance, we consider three types of imbalances simultaneously to construct the loss function.

The following content will comprehensively summarize the problems faced by RSSCD and propose solutions to some of these problems in this article.

1) In multitemporal remote sensing images, there are a large number of nonchanging regions with high differences and changing regions with low differences. The former is prone to false detection, while the latter is prone to missed detection. When the elevation information of each semantic region in the images is missing, it is very difficult to exclude the nonchanging regions with high differences and locate the changing regions with low differences just by relying on the image color information and the 2-D positional information between regions. Moreover, most of the current RSSCD datasets only provide bitemporal data, while the actual changes in land semantic coverage categories are usually progressive and continuous. Therefore, in this article, the method of frame interpolation is adopted to obtain a multitemporal frame sequence from the bitemporal images. This sequence reflects the change process of the land semantic coverage categories. At the same time, taking into account the data complexity, this article conducts global feature interaction and enhancement on the data obtained after scanning along the three dimensions of length, width, and time based on Mamba [9], realizes the extraction of more information, and ultimately improves the model's ability to locate change and exclude nonchange.
2) There are semantic change regions of various scales in multitemporal remote sensing images, and it is difficult for current models to fully capture the changes of all scales. At the same time, the oblique change regions in remote sensing images with a bird's-eye view are also more difficult to locate compared with the horizontal and vertical change regions. In this article, the fineness of the receptive field is increased by obtaining shallow features with a larger receptive field, so as to fully capture multiscale (MS) changes. At the same time, based on the method for solving the first problem, this article designs additional data scanning directions, respectively, for bitemporal remote sensing images and their frame interpolation results. While enhancing the ability to perceive the information interaction between features, this design improves the model's ability to locate the oblique change regions. Ultimately, it helps the model detect change regions distributed on multiple scales and in more directions.
3) There is a serious class imbalance situation in the data of RSSCD, which can be generally divided into three types. First, there is an order-of-magnitude gap in the number of changed and unchanged pixels. Second, the numbers of various ground coverage categories are uneven. Third, there are significant gaps in the number of pixels in various categories of semantic change. To this end, this article proposes consistency dynamic weighted loss (CDW Loss). The first is the consistent loss function, and the second is the dynamic weighting parameter. The former adds constraints to the semantic segmentation results of bitemporal images in nonchanging regions. By doing so, it ensures the consistency of these results and indirectly helps to alleviate various class imbalance problems. The latter dynamically adjusts the weight ratio of the loss results of each part according to the size of the changing regions in the data, the number of various ground coverage categories, and the proportion of each semantic change region, helping the model improve its ability to deal with various class imbalances.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

The subsequent structure of this article is arranged as follows. Section II is about the related work on RSSCD. Section III elaborates in detail on the structure of this article and the calculation formula of the proposed loss function. The experimental results along with the corresponding discussions are presented in Section IV. Finally, Section V summarizes the findings and draws the conclusion of this article.

II. Related Work

A. RSSS

RSSS is a task that obtains the category of the land cover in each pixel belongs by analyzing information, such as the color, texture, shape, and spatial relationships, of a single input image. First, Huang et al. [10] proposed that, compared with the semantic segmentation task of natural scenes, the RSSS model needs to deal with a large number of unique problems of remote sensing images acquired from an aerial perspective: first, the interclass differences of different land cover categories are small, but the intraclass differences are large; second, at the same resolution, there are often huge scale differences among various regions or objects, such as airports and airplanes; and third, whether calculated based on the number of regions or the area, there is a serious imbalance among different land cover categories in remote sensing images.

To address the first problem, many methods start from modeling the information at various scales of the images and achieving the fusion of global and local information. Therefore, most of the current models construct an encoder–decoder architecture based on U-Net [11]. Such an architecture can enable the transfer of information between features at adjacent scales, allowing for multilevel modeling and sufficient contextual fusion. Yue et al. [12] and Li et al. [13], on the basis of U-Net, have achieved a more sufficient interaction and fusion of feature information from different perspectives by adding various additional connections to the model or making appropriate deformations. However, the above-mentioned methods usually introduce noise in low-level features into high-level features because they do not consider the inherent differences between features at different levels. To address this, Zhao et al. [14] have utilized the self-attention mechanism, the gating mechanism [15], and the deformable convolution [16] to achieve the matching of low-level and high-level features, and then fuse them to obtain more prominent features for the generation of the final segmentation results.

In addition to considering the interactive modeling of spatial MS contextual information, Li et al. [17] have overcome the problem of high interclass similarity but low intraclass similarity by obtaining more abstract global information of the input images and more refined class-level context. At the same time, it is also an important measure to assist RSSS by using other related tasks, such as boundary detection, super-resolution image generation, and change detection. The models obtained in this way have better guarantees in terms of generalization and robustness. Zhang et al. [18] have explored the method of combining the boundary detection task, which has the highest compatibility with RSSS.

Regarding the problem of large-scale differences among various targets or semantic regions in remote sensing images, Zheng et al. [19] designed a specific foreground perception relationship network to provide high-quality foreground modeling. Ma et al. [20] implemented foreground attention through a foreground activation branch based on the feature pyramid. Xu et al. [21] proposed the RSSFormer (a remote sensing segmentation framework) to adaptively suppress background noise, thereby highlighting foreground targets, especially small targets.

Next is the problem of class imbalance existing in the RSSS dataset. Lv et al. [22] jointly learned the feature extractor and the classifier head in the first stage. Then, with the fixed representation, they relearned the classifier head by using the class-balancing strategy in the second stage. Zhong et al. [23] proposed a two-stage method that simulates the dependencies between pixels through regional rebalancing and neural collapse, which also has certain reference significance for RSSS.

B. Remote Sensing Binary Change Detection

RSBCD is an application-oriented task that generates binary change results by analyzing multitemporal remote sensing images of the same geographical location at different times. There are three main problems in this field: first, how to obtain an appropriate binary change detection dataset as the basis for the training and optimization of model methods; second, how to process the input multitemporal images (generally bitemporal images), specifically, how to deal with the common factors of the two images due to being from the same geographical location and the characteristic factors generated due to being collected at different time points; and third, how to fuse the multi-input features to generate binary change detection results.

To address the first problem, the commonly used RSBCD dataset LEVIR-CD, which was produced by researchers, such as Chen and Shi [24], is presented. Regarding the multi-input problem of RSBCD, most models are improvements based on the encoder–decoder architecture. The possible solutions listed by Zagoruyko and Komodakis [25] are all variations of the encoder, including the Siamese network, pseudo-Siamese network, and two-channel network. Zhang et al. [18] considered using randomly synthesized images to pretrain the model with the expectation that the model can learn the meaning of "change." Wang et al. [26] and Han et al. [27] explored the paradigm of first identifying where the change has occurred and then generating the fine regional boundaries within the located regular change areas.

C. Remote Sensing Semantic Change Detection

The current RSSCD method has moved away from the stage of directly using traditional image processing methods, such as change vector analysis (CVA), principal component analysis (PCA), and multivariate alteration detection (MAD) to obtain results on raw image data [3]. On the contrary, with the gradual evolution of deep learning (DL) technology and the rapid enrichment of computing resources, the application of DL

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

technologies, such as convolution neural network (CNN), self-attention mechanism, and state space model (SSM) for image feature extraction and result generation has become the main way to solve the RSSCD problem. The following research on RSSCD focuses on DL-based methods.

Daudt et al. [1] proposed four feasible task strategies for RSSCD as shown in Fig. 1. Graves et al. [2] introduced the RNN into the RSSCD to associate multiple remote sensing images with time series relationships and obtain the features of multitemporal images along the time dimension [3]. Peng et al. [6] explored the issue of spatial correlation of the input image pairs and how to deal with the problem of class imbalance in the dataset. Yang et al. [28] adopted an asymmetric architecture to address the problem of inconsistent distributions of input images at two acquisition time points. Ding et al. [29] dealt with RSSCD by increasing the correlation between the segmentation branch and the change branch by enhancing the semantic correlation in time and space and introducing the self-attention mechanism. Wijenayake et al. [30] proposed a joint spatiotemporal fusion method based on Mamba, which alleviates the impact of class imbalance. Wang et al. [31] developed a method that balances the extraction of fine-grained local details with the integration of global contextual information, effectively preserving local information. Xu et al. [32] designed a multitask joint optimization method to optimize the main task of semantic change detection. Meanwhile, Tian et al. [33] explored the approach of first localizing and then detecting, which is also a feasible research direction to improve the acuity of the features of the areas with possible changes.

III. Methodology

A. Preliminaries

1) Target of RSSCD: RSSCD is a compound task. Its input consists of multitemporal remote sensing images of the same location acquired at different times. Through pixel-by-pixel analysis, RSSCD outputs the change situations at various positions in the images. When a change area is detected, RSSCD further identifies the land semantic categories. Assuming that there are n land semantic coverage categories in the dataset, RSSCD is essentially a pixel-level classification task with $n^2 + 1$ categories. There are three paradigms from input to output, referring to the strategy of the common RSSCD model in Fig. 1. First, the model based on the strategy in Fig. 1(a) obtains the final semantic change detection result. It does this by generating the semantic segmentation results of the bitemporal images, respectively. Assuming results of RSSS in two images are LCM_{t_1} and LCM_{t_2}, and result of RSBCD is M, the formula of RSSCD is as follows:

$$SCD = \left(\left(LCM_{t_1} - 1\right) * n + LCM_{t_2}\right) * M. \tag{1}$$

It should be noted that for RSSCD, in addition to detecting the changes in areas with different land semantic coverage categories, if the semantic representation is the same but the area visually changes, such as a house being demolished and then rebuilt, it is still considered a case of change. Moreover, for the areas where the semantic change detection results calculated according to (1) are multiples of 10, they all represent the changes within the same type of land semantic coverage categories. The models used in this article are all based on the strategy shown in Fig. 1(d).

2) Theoretical Basis of Mamba: The RSSCD model in this article is obtained by improving Vmamba, and Vmamba derives from migrating Mamba from sequential tasks to the field of digital image processing. This section will introduce the concept and theoretical basis of Mamba. The Mamba stems from the SSMs [9]. It is a linear model. This model is used to describe the state representation and predict the probability of the next state based on the current model input. SSM can be divided into two parts: one is the state equation, and the other is the output equation. The formulas are as follows:

$$h'(t) = Ah(t) + Bx(t) \tag{2}$$

$$y(t) = Ch(t) + Dx(t). \tag{3}$$

Formula (2) describes the influence of the current state and the model input on the state change. On the other hand, (3) describes how to calculate the model output from the current state and the model input. SSM is a model that obtains continuous-time output from continuous-time input. In order to transform the discrete input into a continuous signal that can be used by SSM, zero-order hold is often used to make the discrete data continuous. This technology keeps the value of each received discrete signal unchanged until a new signal is received. After having continuous signal input, SSM can generate a continuous output. Then, the discretized output can be obtained. After applying the zero-order hold technology, the matrices A and B can be discretized according to the following formulas:

$$\bar{A} = e^{\Delta A} \tag{4}$$

$$\bar{B} = (\Delta A)^{-1}\left(e^{\Delta A} - I\right)\Delta B. \tag{5}$$

The formulas of the discretized SSM model are as follows:

$$h_k = \bar{A}h_{k-1} + \bar{B}x_k \tag{6}$$

$$y_k = Ch_k. \tag{7}$$

After obtaining the calculation method for the input-to-output in a single instance, there is a way to get the outputs at all moments. As long as (6) and (7) are cyclically called for calculation starting from the first moment, the outputs at all moments can be gradually obtained. However, the linear calculation characteristic of this structure makes its calculation efficiency relatively low during the training and inference processes. To this end, an acceleration operation strategy for 1-D convolution can be obtained by referring to the way of using 2-D convolution for parallel operations in image tasks, as shown in the following formulas:

$$\bar{K} = \left(C\bar{B}, C\bar{A}\bar{B}, \ldots, C\bar{A}^{L_x-1}\bar{B}\right) \tag{8}$$

$$y = x * \bar{K}. \tag{9}$$

In the above two formulas, L_x represents the length of the input, and the symbol "*" represents the convolution operation.

The structured state spaces for sequences (S4) are linear time-invariant (LTI) SSMs. This model uses high-order polynomial projection operators (HiPPOs) [34] to assist matrix A

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

in the SSM to retain more contextual information. HiPPO will compress all the input signals into a coefficient vector, which is used to construct matrix A that can well retain historical information, the formula is as follows:

$$A_{nk} = -\begin{cases} \sqrt{(2n+1)(2k+1)}, & n > k \\ \dfrac{1}{n+1}, & n = k \\ 0, & n < k. \end{cases} \tag{10}$$

The matrix A, constructed based on HiPPO, can achieve better performance compared with direct randomization of parameters, and it is capable of reconstructing new signals more accurately.

Mamba is a new SSM architecture based on S4, which has two main improvements. First, it incorporates the selective scan algorithm, which allows the model to focus more on important information by filtering out irrelevant information. Second, Mamba applies an innovative hardware-aware algorithm to improve the model's running speed. Due to the first improvement, Mamba gains the ability to compress small-scale but useful state information into the state matrix. Due to the linear time-invariant characteristics of SSM and S4, matrices A–C are the same for each input. That is to say, they are static models and cannot perceive the specific content of the input. However, Mamba has the ability to selectively focus on the input. Take text sequence tasks as an example. In such tasks, it will selectively ignore punctuation marks and some conjunctions, and instead focus on more critical nouns and verbs. Moreover, Mamba can establish associations between the current input and distant inputs rather than overly relying on adjacent context information. Regarding the new hardware-aware algorithm, Mamba draws inspiration from flash attention [35] and breaks the bottleneck in time consumption by restricting the information transmission between dynamic random access memory (DRAM) and static random access memory (SRAM). Specifically, it retains the parameter backtracking of matrices A–C in DRAM while keeping h in SRAM.

In conclusion, due to Mamba's characteristic of global context aggregation with linear complexity, it has a receptive field for global data compared with CNN. Moreover, compared with Transformer, it has obvious advantages in terms of computational complexity. Mamba not only performs excellently in sequence tasks, such as text and speech, but can also be applied to image-related tasks.

3) Image Processing Method Based on Mamba: Remote sensing image data belongs to 2-D data. If we want to put them into Mamba-related models for global context information collection, we need to perform linearization on the data. Some researchers have explored applications in this area. For example, Vision Mamba [36] and Vmamba [36] have a performance comparable to that of the most advanced CNN models and Transformer-based models. The model in this article is improved from Vmamba. Next, we will introduce how Vmamba realizes the one-dimensionalization of 2-D image data, how to use Mamba for information extraction, and how to generate results according to the task objectives. The core of Vmamba is to stack the visual state-space (VSS) blocks based on the 2-D selective scan (SS2D) module. The SS2D module achieves the one-dimensionalization of 2-D features using multiple scanning directions. As shown below in Fig. 3, there are a total of four scanning directions, namely horizontal sequential scanning, horizontal reverse scanning, vertical sequential scanning, and vertical reverse scanning.

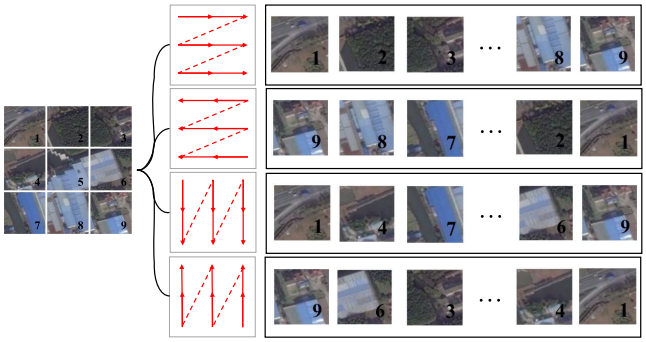


Fig. 3. Four scanning directions of SS2D.

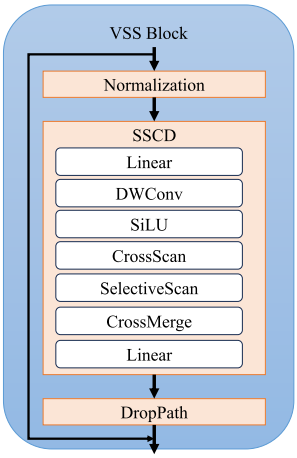


Fig. 4. VSS block.

After completing the 1-D scanning of the data, the SS2D module uses the selective structured state-space sequence model (S6) module to obtain the corresponding output from the features through the state-space sequence. Then, it restores the serialized features to a 2-D structure along the scanning direction according to the previous scanning order. Finally, the results of the four scanning directions are fused as the output of the SS2D. Therefore, the choice of scanning method has a crucial impact on the performance of the SS2D. For any position on the 2-D feature map, the closer the scanning order is to it, the stronger the correlation and the more sufficient the semantic interaction.

Next, the VSS block module is constructed based on the SS2D. This module includes an FCL before the input and after the output, a depthwise convolution layer, DropPath [37], as well as some activation layers and normalization layers. Meanwhile, in order to prevent abnormal situations, such as overfitting and gradient explosion, the VSS block module also applies residual connection [38]. Its overall structure is shown in Fig. 4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

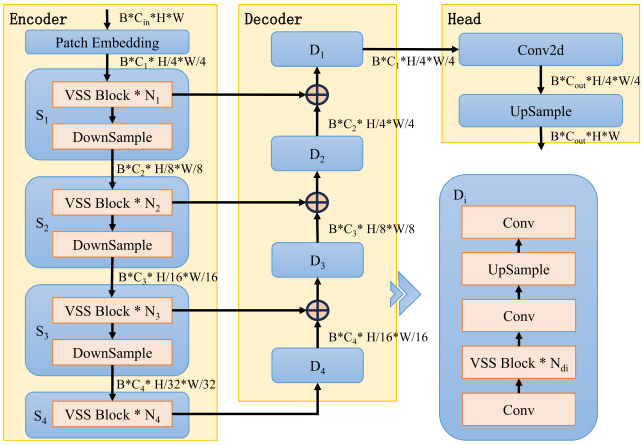


Fig. 5. Image processing basic model based on Mamba.

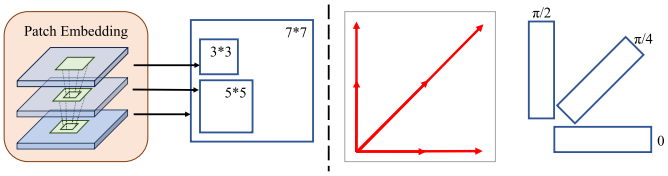


Fig. 6. Our method simulates MS and multidirectional anchors.

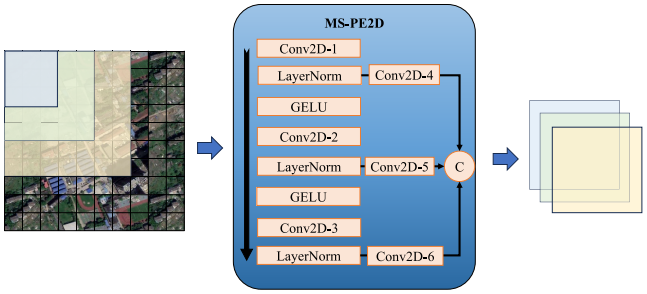


Fig. 7. Structure of MS-PE2D.

Based on this, the basic model for the image processing task based on Mamba is presented, as shown in Fig. 5.

B. Method for Serializing Bitemporal Images

In this article, we need to obtain the temporal dimension information. This information corresponds to the surface location and is obtained from the frame sequence. Then, it combines this temporal dimension information with the 2-D spatial dimension information to generate comprehensive spatiotemporal dimension information. Thus, the first step is to obtain the frame sequence from the image pair. If an image sequence containing N_f frames is constructed in total, then the generation formula of the ith $(0 \le \mathrm{i} < N_f)$ frame is as shown in the following equation:

$$F_i = \Phi\left(I_1, I_2, i\right). \tag{11}$$

In the above formula, I_1 and I_2, respectively, represent the remote sensing images of phase 1 and phase 2 at the same location but at different times. Φ is the frame sequence generation method, which has two boundary constraints. At the same time, we choose to use linear interpolation to generate the frame sequence so as to simulate the intermediate change process between the bitemporal images. Relevant research [39] has proved that the format of reconstructing the input bitemporal images is more important than restoring the real change process. Therefore, (11) can be rewritten as follows:

$$F_i = I_1 + \frac{1}{N_f - 1}\left(I_2 - I_1\right). \tag{12}$$

C. Network Architecture

In this section, the overall structure of the RSSCD model based on Mamba for multidimensional spatiotemporal scanning will be presented, and the functions of each part and the important improvements will be introduced. The structure of the method in this article is shown in Fig. 2, which is generally similar to the fourth strategy of HRSCD shown in Fig. 1(d). The entire model is divided into two major parts. The first part is the semantic segmentation branch, which consists of an encoder, a decoder, and a head. This branch is used to process a single remote sensing image for the extraction and enhancement of spatial information. The second part is the binary change detection branch, which is composed of an encoder, a decoder, and a binary change detection head. This branch is responsible for processing the generated pseudovideo frame sequence to extract and enhance spatiotemporal information.

First, the semantic segmentation encoder will, respectively, obtain the features with the size of 1/4, 1/8, 1/16, and 1/32 of the original length and width from the two input remote sensing images. In order to perceive change areas with more scales, we select to obtain features with more scales from patch embedding to simulate the different sizes of anchor boxes, as shown in Fig. 6, referring to the operation of predefining anchor boxes in the two-stage method of the rotated object detection task [40].

The original patch embedding module can only output a result where the length and width become 1/4 of the original size after two convolutions. In the output features, the receptive field of each feature point in the 2-D position is 5×5. After being processed by the encoder of the semantic segmentation branch, this set of features undergoes a transformation. It is transformed into features of four scales, which are then used by the decoder to generate the required results. Actually, in each submodule of the encoder, there are no other operations that change the feature scale except for the convolution used for downsampling. However, such a patch embedding can only generate features of four scales, corresponding to four receptive field sizes. To address this issue, our method will modify the patch embedding, referring to relevant research [41], so that its output features can have receptive fields of more sizes. As shown in Fig. 7 below, it is the modified MS 2-D patch embedding (MS-PE2D) module.

In Fig. 7, C represents the concatenation operation along the channel dimension. The kernel size of all 2-D convolutions in MS-PE2D is 3×3. The strides along the length and width

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

TABLE I

TABLE COMPARING THE OUTPUT FEATURE RECEPTIVE FIELDS OF THE ENCODER AT VARIOUS SCALES IN PE2D BEFORE AND AFTER IMPROVEMENT

Module name	Feature scale	RF size
PE2D	H/4 * W/4	7*7
	H/8 * W/8	15*15
	H/16 * W/16	31*31
	H/32 * W/32	63*63
MS-PE2D	H/4 * W/4	7*7/11*11/15*15
	H/8 * W/8	15*15/19*19/23*23
	H/16 * W/16	31*31/35*35/39*39
	H/32 * W/32	63*63/67*67/71*71

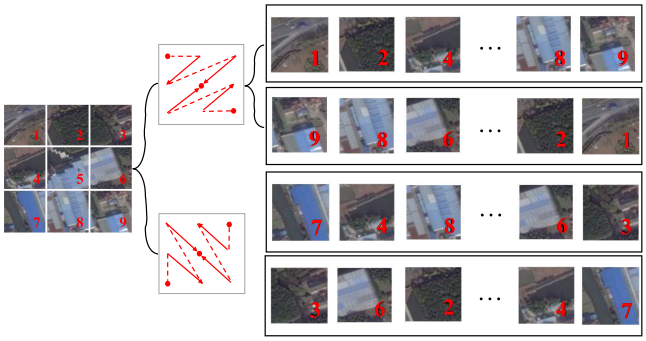


Fig. 8. Added 2-D scanning method.

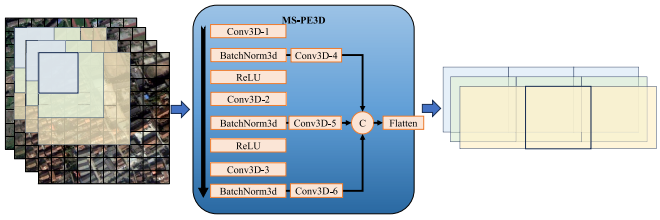


Fig. 9. Structure of MS-PE3D.

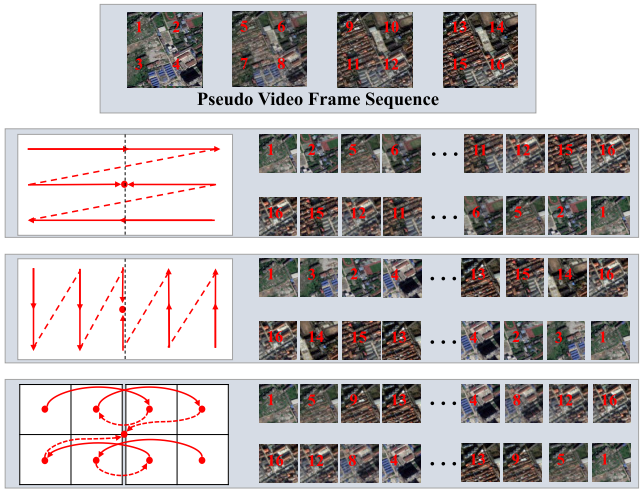


Fig. 10. Six scanning directions of SS3D.

dimensions of Conv2D-1 and Conv2D-2 are 1, and the strides of the other four convolutions are 2. As shown in (13), the change of the receptive field of features after each convolution can be calculated, where RF_n represents the receptive field of the nth layer of convolution, f_n represents the kernel size of the nth layer of convolution, and s_i represents the stride size of the ith layer

$$RF_n = RF_{n-1} + \left[(k_n - 1) \times \prod_{i=1}^{n-1} s_i \right]. \tag{13}$$

According to the above formula, the size of the receptive field contained in each scale output of the encoder before and after the improvement of patch embedding can be calculated, as shown in Table I.

Next, each submodule of the semantic segmentation encoder and decoder employs a more diverse 1-D scanning method for 2-D data compared to that in Fig. 3. Since adjacent information during scanning will have more comprehensive interaction during subsequent calculations, there is a situation. The result of scanning in Fig. 3 often pays more attention to the information in the same row or the same column for the features at various positions in the 2-D space. Consequently, it is more sensitive to the change regions where the direction distribution is perpendicular to the length and width. However, the information at each position should also interact with the diagonally adjacent features. This not only enhances the model's ability to perceive changes in various direction distributions but also improves the accuracy when refining the boundaries of the change regions. The added scanning method is shown in Fig. 8.

Then, we will introduce the structure of the change detection branch. As a whole, it is the same as the semantic segmentation branch, except for the 3-D patch embedding (PE3D) and the 3-D selective scan (SS3D). The structure of the former is shown in Fig. 9. The final flatten operation unfolds, along the W dimension, the result after 3-D convolution to eliminate the time dimension T and achieve two-dimensionalization of the data. The latter conducts parallel scanning of the frame sequence features in the temporal and spatial dimensions along six directions.

Assume that the length of the frame sequence is 4, and the size of the image after passing through the PE3D is 2×2. As shown in Fig. 10, there are six 3-D data scanning directions for the binary change detection branch. When performing temporal dimension scanning in the VSS block, the model first traverses the feature data of all time points at the same position, and then moves the position of interest to the next position in the 2-D spatial order from left to right and top to bottom. The spatial dimension scanning is carried out according to the four-directional scanning method in Fig. 3 after the frame sequence is stitched into a rectangular overall image along the width dimension.

In the binary change detection branch, there are some operations. In addition to adding the same four-directional spatial scanning as in the semantic segmentation branch, when transforming the data in the time dimension to a higher dimensionality, the temporal information will first be aggregated along the diagonal direction. This is done before moving the spatial position. That is, a total of six scanning

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

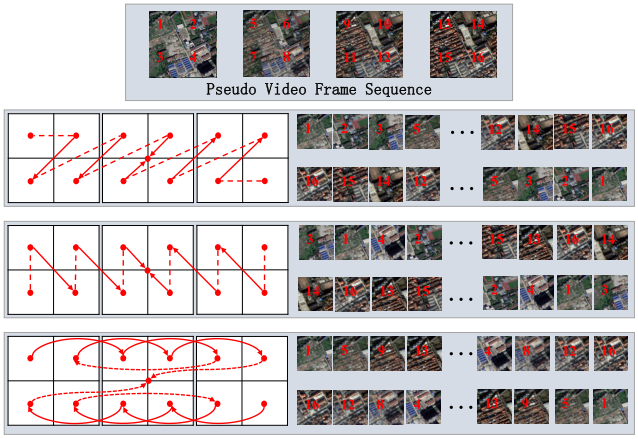


Fig. 11. Added 3-D scanning method.

methods are added. The added scanning methods are shown in Fig. 11.

Finally, the supervision head of the change detection encoder is used to help the model to deal with possible situations, such as overfitting and gradient explosion.

D. Consistency Dynamic Weighted Loss

The commonly used loss function for semantic change detection, focal loss, can only calculate the classification errors on a pixel-by-pixel basis. However, this method is inconsistent with the IoU index based on the intersection and union of regions. Therefore, the loss function in this article introduces Lovasz–Softmax loss [42] to fill the gap of focal loss in regional-level loss calculation.

The loss function proposed in this article mainly consists of two parts. One is the consistency loss (CL) function, and the other is the weighted dynamic loss function, as shown in the following equation:

$$L = L_{Cons} + L_{WD}. \tag{14}$$

For a task where the input has a width and height of W and H, respectively, and the target contains C classes, the calculation of L_{Cons} can be shown in the following equation:

$$L_{Cons} = \sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{C} \left\{\left[p_{s1}(i,j,k) - p_{s2}(i,j,k)\right] * \left(1 - p_{bcd}(i,j)\right)^2\right\}. \tag{15}$$

In the above formula, $p_{s1}(k,i,j)$ and $p_{s2}(k,i,j)$ represent the probabilities that the pixel at the position of the ith row and jth column in the two-temporal remote sensing image belongs to the kth class. These probabilities range between 0 and 1. $p_{bcd}(i,j)$ is the result predicted by the binary change detection branch indicating whether a change has occurred at the corresponding position, where 1 represents a change and 0 represents no change. Besides, by taking the negation of $p_{bcd}(i,j)$ and adding 1, the mask of the unchanged positions can be obtained. L_{Cons} can help the model maintain the consistency of the results of the two-temporal images in the areas output by the semantic segmentation branch that remain unchanged.

The dynamic weighted loss L_{WD} consists of three parts, which can be shown in the following equation:

$$L_{WD} = L_{bcd} + L_{ss} + L_{scd} \tag{16}$$

where L_{bcd} is the binary change detection loss function, L_{ss} is the semantic segmentation loss function, and L_{scd} is the semantic change detection loss function. Their calculation formulas are shown in the following equations:

$$L_{bcd} = \sum_{k=0}^{1} \omega_k \left(L_{bcd_{Lov}} + L_{bcd_{FL}} + L_{pv_{Lov}} + L_{pv_{FL}}\right) \tag{17}$$

$$L_{ss} = \sum_{k=0}^{N} \omega_k \left(L_{s1_{Lov}} + L_{s1_{FL}} + L_{s2_{Lov}} + L_{s2_{FL}}\right) \tag{18}$$

$$L_{scd} = \sum_{k=0}^{N^2} \omega_k \left(L_{scd_{Lov}} + L_{scd_{FL}}\right). \tag{19}$$

In (17), $L_{bcd_{Lov}}$ and $L_{bcd_{FL}}$ are the loss values calculated by using the Lovasz–Softmax loss, and FOCAL LOSS functions, respectively, for the output of the binary change detection branch and the corresponding annotations. $L_{pv_{Lov}}$ and $L_{pv_{FL}}$ are the loss calculation results of the encoder supervision head of the change branch and the ground truth. In (18), $L_{s1_{Lov}}$ and $L_{s1_{FL}}$ are the losses calculated by using the Lovasz–Softmax loss and focal loss functions, respectively, between the semantic segmentation results obtained from the remote sensing image of the first temporal phase and the ground truth annotations. The same applies to $L_{s2_{Lov}}$ and $L_{s2_{FL}}$. In (19), $L_{scd_{Lov}}$ and $L_{scd_{FL}}$ are the loss values calculated by using the Lovasz–Softmax loss and focal loss functions.

In the above formulas, the weight ω_k corresponding to any category is determined by its area proportion, that is, the proportion r_k of the number of pixels. Assuming there are a total of C categories, (20) and (21) are used to show the calculation formulas for the weight ω_k of the kth category, as follows:

$$\omega_k = \frac{f(r_k)}{\sum_{i=1}^{C} f(r_i)} \tag{20}$$

$$f(r_i) = \frac{1}{ln(1 + r_i)}. \tag{21}$$

In this article, the logarithmic inverse function as shown in (20) is used to reduce the gap in the proportion of each category, and then normalization is carried out through (21) to obtain the loss weight of each category.

IV. Experiments and Discussions

A. Datasets

1) Landsat–SCD [43]: This dataset is created from the data collected by Landsat satellites between 1990 and 2020. Its main area of focus is Tumxuk City, Xinjiang, China. It contains 8468 pairs of images with a fixed size of 416×416 and a resolution of 30 m. In terms of the types of land cover, besides the unlabeled class, Landsat–SCD also includes four categories: farmland, desert, buildings, and water surfaces.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

Due to the image resolution, the areas of the buildings in the Landsat–SCD are usually small and scattered. The labeled areas of each category generally have high connectivity. At the same time, approximately 19% of the areas in the entire dataset have experienced changes in land cover types. The Landsat–SCD can effectively test the adaptability of models to remote sensing images with relatively low resolution. Although many current researchers are pursuing models that can better handle high-resolution and ultrahigh-resolution images, remote sensing images with medium and low resolution, especially historical images taken ten years or even decades ago, can still provide a great deal of assistance to relevant research work. Moreover, the distribution ratio of changed and unchanged pixels in this dataset is relatively uniform, making it suitable to serve as a positive sample when exploring the sensitivity of models to the uniformity of data distribution.

2) SECOND [28]: This dataset is currently one of the most commonly used datasets in the field of RSSCD. It is a high-resolution remote sensing image semantic change detection dataset. Compared with the Landsat–SCD, it focuses on the changes in the same land semantic coverage category on the Earth's surface, that is, the intraclass changes. The entire dataset consists of 4662 pairs of bitemporal remote sensing images, and it focuses on the locations on the Earth's surface, which are concentrated in the urban and suburban areas in Hangzhou, Chengdu, and Shanghai. The images in this dataset have a size of 512×512, with a resolution ranging from 0.5 to 3 m and pixel-level annotations. At the same time, the SECOND dataset has been evaluated and calibrated by relevant experts, and it has an extremely high accuracy of ground truth. The entire dataset contains six land cover categories, namely nonvegetation surface, trees, low vegetation, water, buildings, and playgrounds. These categories can be generally divided into two major categories: natural surface and artificial surface. Theoretically, changes between any two land cover categories should occur in the dataset. However, in reality, semantic changes between water and playgrounds do not exist, and changes from water to water also cannot be annotated. Therefore, whether in the training set or the test set, the SECOND cannot contain all possible semantic changes. Therefore, when using the SECOND dataset for method evaluation, it is usually divided into a training set with 2968 pairs and a test set with 1694 pairs according to the dataset division strategy.

3) Hi-UCD [44]: This dataset is an ultrahigh-resolution remote sensing image semantic change detection dataset. It mainly focuses on the changes in the land cover categories of cities. The image coverage area is mainly Tallinn, the capital of Estonia. It spans an area of 102 km^2 in space and focuses on the changes in this area from 2018 to 2019. At the same time, the training set and the test set of Hi-UCD are in different geographical regions, and there is no overlap caused by cropping. All images have a size of 512×512 and a resolution of 0.1 m. This dataset contains nine land cover categories, which include water, grassland, buildings, greenhouses, roads, bridges, bare land, forest land, and other categories. In addition, it also includes unlabeled areas. Due to its large coverage, ultrahigh resolution, and fine semantic annotation, Hi-UCD has two problems. One is class imbalance, and the other is a large amount of interference noise. The former is mainly due to the large amount of data in this dataset and the short interval between the collection of bitemporal images, leading to imbalances in changed and unchanged pixels, various land coverage categories, and various semantic change categories. The latter is that the ultrahigh image resolution magnifies many targets that were originally not significant from the remote sensing perspective, such as vehicles, shadows, and occlusions. At the same time, the excessively high resolution also increases the difficulty of image preprocessing. Various distortions brought to remote sensing images by radiation differences, solar elevation angles and satellite orbital altitudes also increase the difficulty of semantic change detection in the Hi-UCD dataset.

B. Metrics

Following the existing setup in [45], we use the four metrics of Kappa, $F1$, mean IoU (mIoU), and separated Kappa (SeK) coefficient, all of which are better with higher values. Their definitions can be referred to [29].

C. Experimental Setting

During the training process, in order to expand the dataset and enhance its generalization ability, four processing strategies, namely random cropping, random flipping, random rotation, and normalization, will be applied to the input bitemporal remote sensing data in the experiment. The corresponding annotation images for semantic segmentation and binary change detection will also undergo synchronous operations (excluding normalization), and the size of the images will be uniformly adjusted to 512×512. At the same time, this article adopts AdamW [46], which is more stable during long-term multicycle training, as the optimizer. The initial learning rate is 10^{-4}, and the weight decay is 5×10^{-4}. Moreover, in order to help the model converge gradually during the optimization process, after each fixed number of epoch training cycles, the learning rate will be decreased according to the following equation:

$$NewLR = BaseLR \times \gamma^{\left\lfloor \frac{CurStep}{StepSize} \right\rfloor}. \tag{22}$$

In (22), the current learning rate NewLR is the product of the initial learning rate BaseLR and the decay coefficient. The value of the decay coefficient is determined by the current step number CurStep and γ. That is, for every additional StepSize rounds of training, the learning rate will decay by an additional factor of γ. All models are uniformly trained for 80 000 rounds. All the experimental results in this article are obtained after training and testing on an RTX 4090 with 24 GB. The framework used is PyTorch2.0.1+cu118.

D. Experimental Results

This article uses pioneering and representative works in the RSSCD as the comparison methods, which are HRSCD [1], ChangeMask [47], ClearSCD [48], ScanNet [45], HGINet [49], DEFO-MTLSCD [50], BT-HRSCD [51], and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RESULTS OF COMPARATIVE EXPERIMENTS ON LANDSAT–SCD

Method	Kappa	F1(%)	IOU_{mean}(%)	SeK
$HRSCD_1$[1]	30.78	63.33	70.21	19.03
$HRSCD_2$[1]	28.74	-	72.93	18.21
$HRSCD_3$[1]	36.57	68.65	76.39	24.51
$HRSCD_4$[1]	36.20	67.53	76.52	24.36
ChangeMask[47]	35.54	68.71	75.27	23.31
ClearSCD[48]	27.97	64.35	71.70	17.60
ScanNet[45]	40.48	73.02	77.21	27.51
HGINet[49]	45.31	75.66	79.10	31.65
DEFO-MTLSCD[50]	45.14	75.63	79.06	31.51
BT-HRSCD[51]	45.78	76.31	79.51	32.23
ChangeMamba[52]	46.06	76.21	79.45	32.38
ours(base)	47.03	76.77	79.84	33.27
ours(with MSOS)	52.07	79.68	81.69	38.10

TABLE III
RESULTS OF IMPROVEMENT FOR VARIOUS METHODS AFTER APPLYING CDW LOSS ON LANDSAT–SCD

Method	Kappa	F1(%)	IOU_{mean}(%)	SeK
$HRSCD_3$[1]	36.57	68.65	76.39	24.51
with CDW Loss	44.07	74.91	78.68	30.60
$HRSCD_4$[1]	36.20	67.53	76.52	24.36
with CDW Loss	42.70	74.05	78.12	29.35
ChangeMask[47]	35.54	68.71	75.27	23.31
with CDW Loss	43.21	74.32	78.23	29.77
ClearSCD[48]	27.97	64.35	71.70	17.60
with CDW Loss	31.91	66.98	73.89	20.46
ScanNet[45]	40.48	73.02	77.21	27.51
with CDW Loss	42.56	74.22	78.02	29.31
HGINet[49]	45.31	75.66	79.10	31.65
with CDW Loss	47.45	76.88	80.01	33.67
DEFO-MTLSCD[50]	45.14	75.63	79.06	31.51
with CDW Loss	48.39	77.46	80.40	34.55
BT-HRSCD[51]	45.78	76.31	79.51	32.23
with CDW Loss	51.12	79.06	81.32	37.15
ChangeMamba[52]	46.06	76.21	79.45	32.38
with CDW Loss	50.56	78.85	81.31	36.68
ours(with MSOS)	52.07	79.68	81.69	38.10
with CDW Loss	54.66	81.15	82.66	40.64

TABLE IV
RESULTS OF COMPARATIVE EXPERIMENTS ON SECOND

Method	Kappa	F1(%)	IOU_{mean}(%)	SeK
$HRSCD_1$[1]	12.96	33.96	48.80	6.70
$HRSCD_2$[1]	11.02	-	59.99	6.16
$HRSCD_3$[1]	20.77	41.58	67.10	12.77
$HRSCD_4$[1]	22.11	42.98	67.77	13.63
ChangeMask[47]	21.82	42.85	67.05	13.34
ClearSCD[48]	17.54	40.31	64.75	10.32
ScanNet[45]	32.48	54.28	72.31	21.23
HGINet[49]	31.58	52.63	72.47	20.66
DEFO-MTLSCD[50]	30.67	51.16	72.61	20.06
BT-HRSCD[51]	33.08	54.01	73.04	21.79
ChangeMamba[52]	32.80	53.65	72.75	21.58
ours(base)	33.58	54.30	73.22	22.18
ours(with MSOS)	34.82	55.40	73.45	22.78

TABLE V
RESULTS OF IMPROVEMENT FOR VARIOUS METHODS AFTER APPLYING CDW LOSS ON SECOND

Method	Kappa	F1(%)	IOU_{mean}(%)	SeK
$HRSCD_3$[1]	20.77	41.58	67.10	12.77
with CDW Loss	26.81	47.36	70.22	17.04
$HRSCD_4$[1]	22.11	42.98	67.77	13.63
with CDW Loss	25.66	47.17	69.07	16.05
ChangeMask[47]	21.82	42.85	67.05	13.34
with CDW Loss	26.38	47.86	70.02	16.70
ClearSCD[48]	17.54	40.31	64.75	10.32
with CDW Loss	22.74	43.28	68.39	14.11
ScanNet[45]	32.48	54.28	72.31	21.23
with CDW Loss	34.55	55.05	73.33	22.93
HGINet[49]	31.58	52.63	72.47	20.66
with CDW Loss	33.79	54.49	73.40	22.35
DEFO-MTLSCD[50]	30.67	51.16	72.61	20.06
with CDW Loss	34.04	54.82	73.46	22.50
BT-HRSCD[51]	33.08	54.01	73.04	21.79
with CDW Loss	34.49	55.11	73.53	22.92
ChangeMamba[52]	32.80	53.65	72.75	21.58
with CDW Loss	34.36	54.88	73.19	22.79
ours(with MSOS)	34.82	55.40	73.45	22.78
with CDW Loss	35.08	55.67	73.60	23.32

ChangeMamba [52], which also uses Mamba as the feature enhancement method. As a forerunner in applying Mamba to the field of change detection, ChangeMamba not only includes research on semantic change detection, but also explores the effectiveness of the new scanning method of Mamba for binary change detection tasks and building damage assessment. The model used in this article is also inspired by ChangeMamba to a certain extent. Next, the experimental results will be presented and specifically analyzed from both quantitative and qualitative aspects.

1) Quantitative Experimental Results: In the quantitative experiment, we will verify the effectiveness of the proposed method and the loss function, respectively, on three datasets with different resolutions. It should be noted that HRSCD1–HRSCD4 in Table II are four variants of the HRSCD model, each corresponding to one of the four strategies shown in Fig. 1.

a) Landsat–SCD: First of all, the results of the method in this article and comparison methods on the Landsat–SCD datasets with focal loss [8] are presented as Table II. Specifically, the method proposed in this article will be divided into the basic method and the method with the addition of MS omnidirectional scanning (MSOS). Notably, the calculation methods of these indicators are the same as those in ChangeMamba [52]. We will mark the optimal indicators in red and the suboptimal indicators in blue.

It can be found that the method in this article outperforms other comparative methods on Landsat–SCD. ChangeMamba is the suboptimal model, which also uses Mamba as the feature enhancement method. Then comes BT-HRSCD. At the same time, the method with MSOS also outperforms the basic method.

Next, we present the improvement of the metrics of the method in this article and various comparative methods before and after applying the CDW Loss, as shown in Table III. It can be found that the CDW Loss shows improvement in all metrics for all models that simultaneously have binary change detection outputs and two-temporal semantic segmentation outputs.

b) SECOND: This section presents the relevant experimental results on the SECOND dataset, as shown in Tables IV and V. It can be found that on the SECOND dataset, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

RESULTS OF COMPARATIVE EXPERIMENTS ON HI-UCD

Method	Kappa	F1(%)	IOU_{mean}(%)	SeK
$HRSCD_1$[1]	7.35	21.64	53.49	3.43
$HRSCD_2$[1]	7.78	-	61.99	4.00
$HRSCD_3$[1]	8.69	27.91	61.94	4.29
$HRSCD_4$[1]	14.21	33.91	65.05	7.43
ChangeMask[47]	19.83	38.51	70.54	11.59
ClearSCD[48]	20.76	40.92	69.87	11.93
ScanNet[45]	4.77	22.44	67.08	2.62
BT-HRSCD[51]	18.95	38.51	69.74	10.92
HGINet[49]	19.65	33.79	70.08	11.28
DEFO-MTLSCD[50]	16.77	37.24	69.55	9.61
ChangeMamba[52]	24.64	45.19	70.28	14.23
ours(base)	27.11	46.94	71.07	15.87
ours(with MSOS)	28.55	48.16	72.54	17.19

TABLE VII

RESULTS OF IMPROVEMENT FOR VARIOUS METHODS AFTER APPLYING CDW LOSS ON HI-UCD

Method	Kappa	F1(%)	IOU_{mean}(%)	SeK
$HRSCD_3$[1]	8.69	27.91	61.94	4.29
with CDW Loss	14.75	32.88	66.86	7.98
$HRSCD_4$[1]	14.21	33.91	65.05	7.43
with CDW Loss	22.42	41.88	70.48	13.01
ChangeMask[47]	19.83	38.51	70.54	11.59
with CDW Loss	22.61	40.89	71.77	13.50
ClearSCD[48]	20.76	40.92	69.87	11.93
with CDW Loss	25.01	45.11	71.19	14.74
ScanNet[45]	4.77	22.44	67.08	2.62
with CDW Loss	11.42	30.76	68.49	6.43
HGINet[49]	19.65	33.79	70.08	11.28
with CDW Loss	11.53	29.95	70.73	11.54
DEFO-MTLSCD[50]	16.77	37.24	69.55	9.61
with CDW Loss	24.03	43.51	70.87	14.02
BT-HRSCD[51]	18.95	38.51	69.74	10.92
with CDW Loss	23.33	42.97	70.90	13.68
ChangeMamba[52]	24.64	45.19	70.28	14.23
with CDW Loss	27.60	47.57	71.45	16.28
ours(with MSOS)	28.55	48.16	72.54	17.19
with CDW Loss	29.96	49.50	72.66	18.08

TABLE VIII

MODEL COMPUTATIONAL COMPLEXITY (GFLOPS), NUMBER OF PARAMETERS (MILLION), AND INFERENCE SPEED (FPS) WITH THE INPUT OF 512×512

Method	GFlops	Params(M)	Fps(pair/s)
$HRSCD_1$[1]	8.02	3.36	3.16
$HRSCD_2$[1]	14.29	6.39	3.59
$HRSCD_3$[1]	42.67	12.77	2.24
$HRSCD_4$[1]	43.69	13.71	3.17
ChangeMask[47]	37.16	2.97	3.12
ClearSCD[48]	20.76	69.87	3.83
ScanNet[45]	264.95	27.90	3.31
HGINet[49]	12.65	27.70	2.10
DEFO-MTLSCD[50]	100.27	26.02	2.30
BT-HRSCD[51]	131.26	33.33	9.44
ChangeMamba[52]	146.70	54.28	3.06
ours(base)	46.15	56.63	3.25
ours(with MSOS)	98.51	104.72	2.99

TABLE IX

DIFFERENCE IN THE NUMBER OF CHANGED REGIONS BETWEEN THE METHOD WITH AND WITHOUT MSOS UNDER VARIOUS SCALES

Scale(pixel)	Landsat-SCD	SECOND	Hi-UCD
0-50	+9972	+7794	+93
50-500	+205	+464	+27
500-1000	+3	+158	+36
1000-5000	+58	+575	+24
5000-10000	+22	+85	+12
>10000	+12	+35	+21

TABLE X

DIFFERENCE IN THE NUMBER OF CHANGED REGIONS BETWEEN THE METHOD WITH AND WITHOUT MSOS UNDER VARIOUS ANGLES

Angle(°)	Landsat-SCD	SECOND	Hi-UCD
0-15	+930	+1469	+3
15-45	+2158	+7968	+47
45-75	+6834	+3684	+144
75-90	+380	+1321	+20

method in this article also achieves improvements in all metrics, and CDW Loss can provide significant metric gains for all comparative method.

c) Hi-UCD: This section presents the relevant experimental results on the Hi-UCD dataset, as shown in Tables VI and VII. We can find that, in the ultrahigh-resolution dataset, the method in this article has a very significant improvement compared with other comparative methods. This is because it can more accurately exclude the nonfocused changes after magnification, such as vehicles and other contents, and locate the changes more precisely.

d) Model efficiency: We report the model efficiency of both the baseline model and the final model, including the number of parameters, computational complexity, and inference speed, as shown in Table VIII. It is important to emphasize that the primary advantage of our model lies in its superior accuracy metrics compared to other models, rather than its computational efficiency. As mentioned in the introduction, the Mamba-based model has the advantage of linear computational complexity. Our understanding is that this advantage allows Mamba-based models to stack more network structures without the concern of significantly slower computation due to the increased complexity.

Based on our model efficiency report, although the computational complexity of our model is roughly double that of the baseline model, its computational efficiency still retains more than 90% of the baseline model's efficiency, demonstrating the high efficiency of the proposed model. Furthermore, when compared to other models, although our model is less efficient than most existing methods, the difference is minimal. Our approach sacrifices a small amount of computational efficiency in exchange for significantly higher accuracy. Considering that the computational efficiency requirements for change detection tasks are relatively low, our approach is indeed a sound choice.

e) Statistics on the number of correctly detected change regions: Finally, we will compare the number of changed regions within various scale ranges and oblique distribution correctly detected by the method with MSOS and the basic method on each dataset. As shown in Table IX, it can be found that the method with MSOS detects a larger number of changed regions within each scale range than the basic method,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

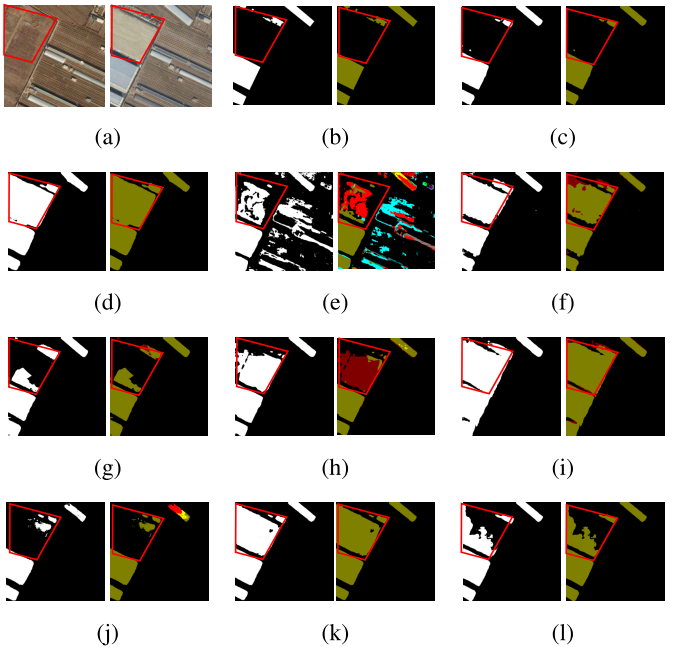


Fig. 12. Comparison of identification results of nonchanging areas prone to false detection. (a) Input. (b) Annotation. (c) Our method. (d) $HRSCD_1$. (e) $HRSCD_2$. (f) $HRSCD_3$. (g) $HRSCD_4$. (h) ChangeMask. (i) ClearSCD. (j) BT-HRSCD. (k) ScanNet. (l) ChangeMamba.

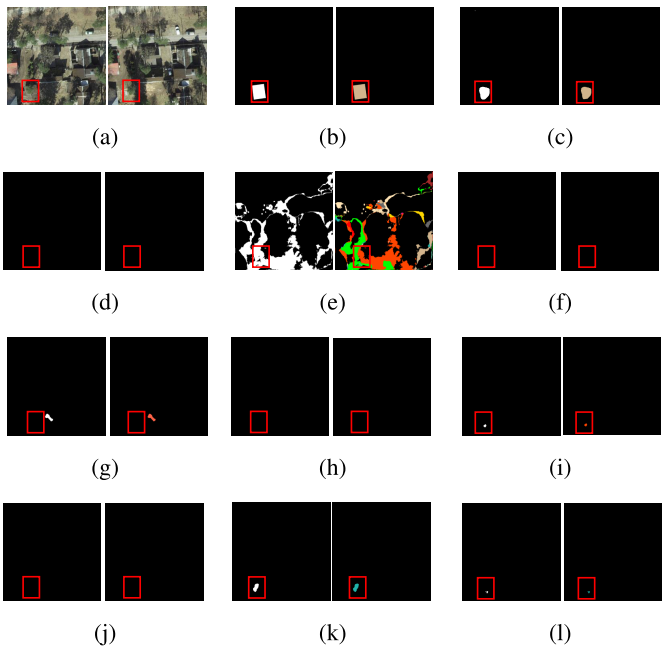


Fig. 13. Comparison of location results of change areas prone to ignore. (a) Input. (b) Annotation. (c) Our method. (d) $HRSCD_1$. (e) $HRSCD_2$. (f) $HRSCD_3$. (g) $HRSCD_4$. (h) ChangeMask. (i) ClearSCD. (j) BT-HRSCD. (k) ScanNet. (l) ChangeMamba.

which proves that it has stronger positioning capabilities for regions of various scale sizes.

Next, we will verify the positioning ability of our method for the areas with oblique distribution changes. The angular distributions of all connected change areas are between 0° and 90°. Table X counts the number difference of change areas correctly detected between the basic method and the method with MSOS within each inclination angle range of each dataset.

2) Qualitative Experimental Results: This section will present a series of qualitative experimental results to demonstrate the effectiveness of the method in our article in addressing issues. Fig. 12 shows the identification results of various methods on nonchange areas that are prone to false detection due to large differences. It can be found that the method in this article has a unique advantage in excluding and identifying nonchange areas. For instance, the bare land in the top-left area of the figure exhibits obvious color differences due to different collection times. Many methods, influenced by the two adjacent and more significant change areas from bare land to buildings in the lower left, tend to directly determine the occurrence of changes in the top left area based on colors. Although there is some slight noise at the lower edge of this nonchange region in our method, as a whole, it has accurately excluded this region, which shows great differences before and after, but has not actually changed.

After that, we will demonstrate the ability of the method in this article to locate insignificant changes, as shown in Fig. 13. There are mainly three reasons for the difficulty in locating the changes in the above figure. First, there is occlusion by trees. Second, there is an interference from the shadows of trees. Finally, the top of the building is covered by fallen leaves, resulting in a very high similarity to the bare land. Our method can precisely help the model to strengthen the association of semantic information at adjacent positions, which gives the model the ability to detect the disappearance of the ground shadow along with the building and to exclude the interference from the surrounding trees and their shadows. Among the comparative methods, only ClearSCD, ScanNet, and ChangeMamba can respond to the possible changes. However, in terms of the positioning accuracy of the change regions, the former three have obvious deficiencies: they not only fail to fully cover the real change regions, but also misclassify the change types.

Finally, we present the qualitative comparison results. These results are about the location capabilities of the method in this article and the comparative methods for MS and oblique change areas. As shown in Fig. 14, it can be found that, with fewer false detections, the method in this article responds to the change areas within all scale and angle ranges. Moreover, the segmentation of the area boundary is more consistent with the real change annotations, and the semantic change classification results are also more accurate.

E. Ablation Study

In this section, we will verify the effectiveness of the change detection encoder supervision head, improving patch embedding (IPE), adding scanning direction, the CL, and the dynamic weighting loss (DWL) in CDW Loss through ablation experiments.

1) Change Detection Encoder Supervision Head: The supervision head is used in our model to assist in the optimization of the binary change detection encoder. Fig. 15 shows the binary change detection results output by the head.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

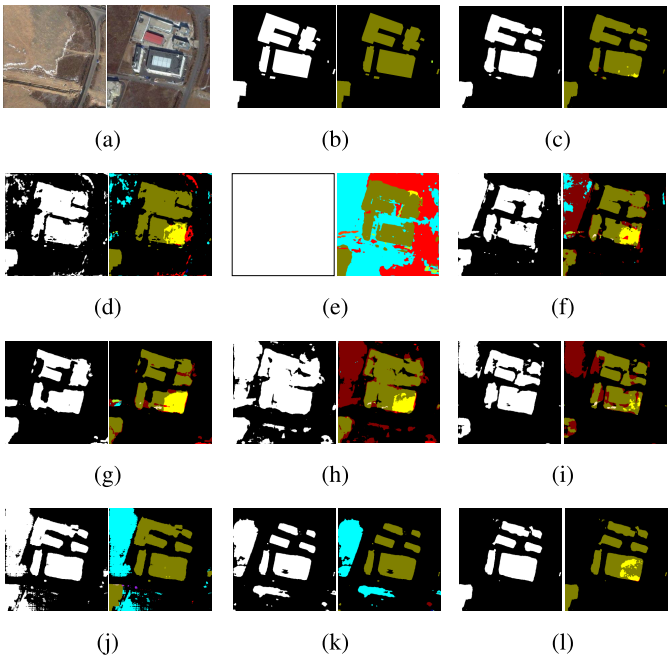


Fig. 14. Comparison of the location results of change areas with different sizes and directional distributions. (a) Input. (b) Annotation. (c) Our method. (d) $HRSCD_1$. (e) $HRSCD_2$. (f) $HRSCD_3$. (g) $HRSCD_4$. (h) ChangeMask. (i) ClearSCD. (j) BT-HRSCD. (k) ScanNet. (l) ChangeMamba.

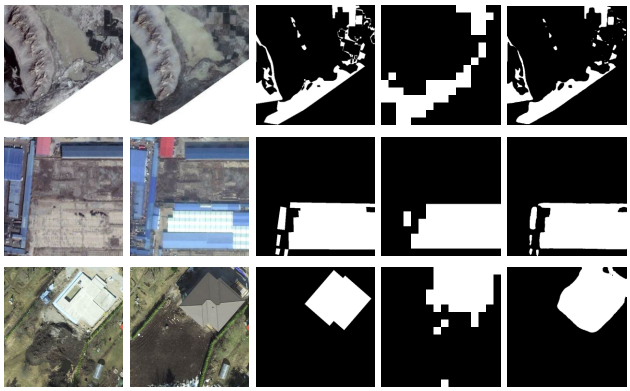


Fig. 15. Visualization Results with the 2-D Scanning Method.

From top to bottom are examples of Landsat–SCD, SECOND, and Hi-UCD, respectively. It can be found that the results have accurately located large-scale changes and small-scale but dense changes. Supervising these results is beneficial to preliminarily determining the areas where changes occur, and helps the method to further refine the shape and contour of the changed areas in the subsequent decoder based on this information.

We have also conducted experiments on three datasets to investigate the impact of the supervision of the binary change detection encoder (with and without it) on the model's indicators. As shown in Table XI, adding the supervision head for the binary change detection encoder effectively improves the accuracy of the method's indicators on datasets with different resolutions, and it is more effective for datasets with higher resolutions. This is because the higher the resolution of the dataset, the more independent the changed areas are, and the relatively larger their occupied areas are. The supervision head happens to have excellent auxiliary recognition ability for large independent changed areas, which can help the method in our article better complete the semantic change detection task in high-resolution data.

TABLE XI

ABLATION EXPERIMENT RESULT ON THE EFFECTIVENESS OF THE SUPERVISION HEAD OF THE BINARY CHANGE DETECTION ENCODER

Method	Dataset	SeK	Δ
w/o Supervision Head	Landsat-SCD	32.84	0
	SECOND	22.17	0
	Hi-UCD	15.32	0
with Supervision Head	Landsat-SCD	33.27	+0.43
	SECOND	22.78	+0.61
	Hi-UCD	16.08	+0.76

TABLE XII

ABLATION STUDY ON THE IPE ACROSS THREE DATASETS

Dataset	IPE	F1 (%)	mIoU (%)	SeK
Landsat-SCD		78.99	81.23	37.02
	✔	79.68(+2.91)	81.69(+1.85)	38.10(+4.83)
SECOND		54.97	73.30	22.66
	✔	55.40(+1.10)	73.45(+0.23)	22.78(+0.60)
Hi-UCD		47.30	72.13	16.45
	✔	48.16(+1.22)	72.54(+1.47)	17.19(+1.32)

TABLE XIII

ABLATION STUDY ON THE SCANNING DIRECTIONS OF SS2D ACROSS THREE DATASETS

Dataset	ASD	F1 (%)	mIoU (%)	SeK
Landsat-SCD		77.52	80.47	34.50
	✔	79.68(+2.16)	81.69(+1.22)	38.10(+3.60)
SECOND		54.93	73.32	22.72
	✔	55.40(+0.47)	73.45(+0.13)	22.78(+0.06)
Hi-UCD		47.15	71.99	16.21
	✔	48.16(+1.01)	72.54(+0.55)	17.19(+0.98)

TABLE XIV

ABLATION STUDY ON THE SCANNING DIRECTIONS OF SS3D ACROSS THREE DATASETS

Dataset	ASD	F1 (%)	mIoU (%)	SeK
Landsat-SCD		77.26	80.09	34.14
	✔	79.68(+2.42)	81.69(+1.60)	38.10(+3.96)
SECOND		54.79	73.26	22.47
	✔	55.40(+0.61)	73.45(+0.19)	22.78(+0.31)
Hi-UCD		47.05	71.58	16.05
	✔	48.16(+1.11)	72.54(+0.96)	17.19(+1.14)

2) Multiscale Patch Embedding and More Scanning Directions: In this section, we will verify the effectiveness of IPE and add the additional scanning directions (ASDs). Table XII shows the results of the ablation experiments of IPE on the three datasets, while Tables XIII and XIV show the ablation experiments of ASD in SS2D and SS3D, respectively, across the three datasets. We can find that both IPE and adding the scanning directions can improve the indicator results based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

TABLE XV
ABLATION EXPERIMENT RESULT OF CDW LOSS ON LANDSAT–SCD

Base	CL	DWL Lov./FL.		F1(%)	IOU_{mean}(%)	SeK
✔				79.68	81.69	38.10
✔	✔			80.76(+1.08)	82.57(+0.88)	40.13(+2.03)
✔		✔		79.99(+0.31)	81.91(+0.22)	38.68(+0.58)
✔			✔	79.94(+0.26)	81.88(+0.19)	38.64(+0.54)
✔	✔	✔		81.07(+1.39)	82.61(+0.92)	40.52(+2.42)
✔	✔		✔	80.76(+1.08)	82.41(+0.72)	39.98(+1.88)
✔		✔	✔	80.38(+0.70)	82.30(+0.61)	39.40(+1.30)
✔	✔	✔	✔	81.15(+1.47)	82.66(+0.97)	40.64(+2.51)

TABLE XVI
ABLATION EXPERIMENT RESULT OF CDW LOSS ON SECOND

Base	CL	DWL Lov./FL.		F1(%)	IOU_{mean}(%)	SeK
✔				55.40	73.45	22.78
✔	✔			55.71(+0.31)	73.47(+0.02)	23.21(+0.43)
✔		✔		55.60(+0.20)	73.47(+0.02)	23.19(+0.41)
✔			✔	55.23(-0.17)	73.48(+0.03)	23.14(+0.36)
✔	✔	✔		55.63(+0.10)	73.58(+0.13)	23.18(+0.40)
✔	✔		✔	55.50(+0.10)	73.55(+0.10)	23.15(+0.37)
✔		✔	✔	55.62(+0.22)	73.48(+0.03)	23.14(+0.36)
✔	✔	✔	✔	55.67(+0.27)	73.56(+0.11)	23.32(+0.54)

TABLE XVII
ABLATION EXPERIMENT RESULT OF CDW LOSS ON HI-UCD

Base	CL	DWL Lov./FL.		F1(%)	IOU_{mean}(%)	SeK
✔				48.16	72.54	17.19
✔	✔			49.32(+1.16)	72.57(+0.03)	17.92(+0.73)
✔		✔		49.16(+1.00)	72.11(-0.43)	17.53(+0.34)
✔			✔	48.93(+0.77)	72.15(-0.39)	17.35(+0.16)
✔	✔	✔		49.42(+1.26)	72.61(+0.07)	17.94(+0.75)
✔	✔		✔	49.23(+1.07)	72.65(+0.11)	17.87(+0.68)
✔		✔	✔	49.22(+1.06)	72.34(-0.20)	17.63(+0.44)
✔	✔	✔	✔	49.50(+1.34)	72.66(+0.12)	18.08(+0.89)

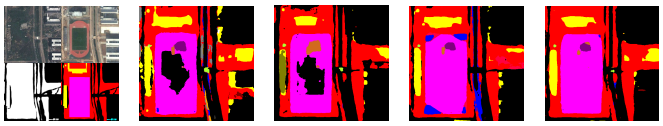
on the basic method. Among them, the effect of the latter in datasets with various resolutions is greater than that of the former in most cases. Thus, it can be seen that the added scanning directions can help the model better locate the changed areas.

From the results presented in Tables XIII and XIV, two clear observations can be drawn. First, the experimental group adopting the baseline SS2D with the improved SS3D consistently outperforms the group using the improved SS2D with the baseline SS3D across all datasets. This indicates that the performance gain brought by the ASD in SS3D (as illustrated in Fig. 8) is greater than that introduced by the ASD in SS2D (as shown in Fig. 11). Considering that the former incorporates a larger number of ASDs than the latter, this outcome is consistent with our expectations. Second, for both SS2D and SS3D, the ASD-enhanced versions clearly outperform their respective baseline models. Based on these findings, we can conclude that introducing ASDs significantly improves the performance of the proposed method, and that the number of scanning directions tends to correlate positively with performance.



(a) (b) (c) (d) (e)

Fig. 16. Visualization comparison of the baseline, DWL-only, CL-only, and DWL and CL settings. (a) Input. (b) Base. (c) DWL. (d) CL. (e) CDWL.



(a) (b) (c) (d) (e)

Fig. 17. Visualization comparison of the baseline, focal loss-only, Lovász loss-only, and DWL settings. (a) Input. (b) Base. (c) FL. (d) Lov. (e) DWL.

3) CL and the DWL in CDW Loss: In this section, we will verify the effectiveness of CL and the DWL in CDW Loss, where DWL consists of two losses: Lovász loss and focal loss. While Tables XV–XVII quantify the improvements, it is crucial to understand the underlying mechanisms governing these gains.

First, it is necessary to analyze the degradation caused by focal loss. A specific anomaly is observed in Table XVI (SECOND dataset), where using focal loss alone degrades performance (−0.17% $F1$). We attribute this to the conflict between the "hard mining" mechanism of focal loss and label noise. The SECOND dataset features complex urban scenarios with potential ambiguities or outliers in annotations. Focal loss forces the model to overfit these noisy "hard" samples, damaging generalization. Lovász loss, with its global averaging property, remains robust to such local noise, thereby avoiding this degradation.

Second, the synergistic interaction between CL and DWL. The combination of CL and DWL consistently yields the best performance, surpassing the sum of their individual contributions. To further clarify their respective contributions, we provide visualized results obtained using only DWL and only CL, as shown in Fig. 16. From the figure, it can be observed that DWL significantly alleviates the false detections caused by the dominant (beige) class, thereby mitigating the class imbalance issue. However, it tends to be overly aggressive, introducing sparse classes (green) that are not actually present in the scene. In contrast, CL produces smoother segmentation boundaries and improves semantic classification performance. When DWL and CL are jointly applied, both the class imbalance problem and the overly aggressive classification behavior are effectively mitigated, leading to the best overall performance. This indicates a complementary interaction: DWL addresses class imbalance by forcing the network to focus on sparse change areas. However, this aggressive reweighting can introduce training instability or increase intraclass variance. CL imposes invariance constraints on the feature space. It acts as a regularizer, smoothing the decision boundaries and correcting feature distribution shifts caused by DWL. Consequently, while DWL enhances the model's ability to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

TABLE XVIII
ABLATION EXPERIMENT OF FRAME INTERPOLATION WITH DIFFERENT FRAME NUMBERS ON LANDSAT–SCD

frames	F1(%)	IOU_{mean}(%)	SeK
2	77.56	80.48	34.70
4(ours)	**79.68**	**81.69**	**38.10**
6	79.59	81.63	37.91
8	79.63	81.69	38.14

detect rare changes, CL ensures the reliability of the learned features. This synergy explains why the negative impact of FL in the SECOND dataset is fully mitigated when combined with CL, as CL effectively alleviates the noise-induced overfitting.

Finally, the contribution of Lovasz and focal losses. Our ablation studies reveal that Lovász loss outperforms focal loss within the DWL module in most cases. This stems from their optimization paradigms: focal loss is strictly pixelwise, treating each pixel independently. While it reweights hard examples, it neglects the geometric structural integrity of the changing objects. In contrast, Lovász loss is region-based, acting as a surrogate for the Jaccard index (IoU). We further report the visual differences when each loss is applied individually, as shown in Fig. 17. When only FL is employed, the misclassification along the edges of the rectangular region corresponding to the prominent hole (blue) is alleviated to some extent, and the area of the hole is significantly reduced. However, additional false positives appear in the bottom-right region. This suggests that, in the proposed framework, FL mitigates the class imbalance problem in a relatively aggressive manner. When only the Lovász loss is used, the hole region is completely filled without introducing noticeable false positives. Nevertheless, the misclassification along the edges of the rectangular region is not significantly improved. This indicates that Lovász loss places greater emphasis on the overall segmentation shape, while paying comparatively less attention to specific semantic categories. When both losses are jointly applied, their complementary strengths are effectively integrated, yielding improved performance in alleviating multiclass imbalance as well as the imbalance between changed and unchanged areas. Since change detection relies heavily on object completeness rather than discrete pixel classification, the global optimization of Lovász loss aligns better with the evaluation metrics, offering superior robustness.

4) Number of Frames in Frame Interpolation: In this section, we will evaluate the impact of different frame counts in the frame interpolation method on the results. While the effectiveness of frame interpolation has been implicitly validated in multimodule collaboration, it is still necessary to test how different frame settings influence the outcomes, further demonstrating the validity of the frame interpolation method. We conducted experiments on Landsat–SCD with frame counts of 2, 4, 6, and 8, as shown in Table XVIII. As can be seen, there is a significant performance improvement from two to four frames. However, for the four-, six-, and eight-frame settings, there is no noticeable difference in performance. This suggests that the number of frames in frame interpolation exhibits a clear edge effect. Beyond four frames, performance no longer improves significantly, and it introduces greater computational complexity.

F. Discussions of Limitations

It is pertinent to discuss the performance characteristics of our proposed model across the three distinct datasets. While our method achieves significant performance gains on the Landsat–SCD and Hi-UCD datasets, its improvement over existing state-of-the-art models on the SECOND dataset is more modest. We attribute this performance profile to the specific characteristics of the datasets themselves and the corresponding strengths of our model's architecture.

The SECOND dataset is a widely adopted benchmark in the RSSCD field, characterized by its high-quality, high-resolution imagery and expert-calibrated, highly accurate ground-truth annotations. As one of the most commonly used datasets, many existing methods have been extensively tested, fine-tuned, and potentially overoptimized for its specific data distribution and change scenarios, which primarily focus on intraclass changes within well-defined urban and suburban areas. This has resulted in a very high-performance ceiling, where achieving substantial improvements becomes incrementally more difficult. We hypothesize that the relatively "clean" nature of the SECOND dataset does not fully showcase the primary advantages of our model, which are more pronounced when dealing with complex data challenges.

In summary, a limitation of our current model is that its competitive advantage is less pronounced on standardized, high-quality benchmarks, like SECOND, where baseline models are already highly optimized. However, this is counterbalanced by its demonstrated superiority in more challenging and realistic scenarios, involving low-resolution historical imagery or noisy, imbalanced ultrahigh-resolution data. Future work could explore hybrid approaches or architectural modifications to further enhance its performance on standard benchmarks without compromising its robustness in complex environments.

Another limitation of our proposed model pertains to its computational efficiency. Compared to some lightweight baseline architectures, our model exhibits a slightly higher computational cost in terms of both GFLOPs and Params. This increased complexity is a direct consequence of several architectural choices made to enhance performance, including the integration of the frame interpolation strategy and the MSOS. While these components are crucial for capturing complex dependencies and achieving superior accuracy, especially on challenging datasets like Hi-UCD, they inherently demand greater computational resources.

This tradeoff between performance and efficiency is a critical consideration. Our primary objective in this study was to push the boundaries of detection accuracy and robustness in complex change detection scenarios. As such, we prioritized model efficacy over computational speed. However, we recognize that for real-time applications or deployment on resource-constrained platforms, the current model's efficiency may pose a challenge. Future research could focus on optimizing this aspect, exploring techniques such as knowledge

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

distillation, model pruning, or designing a more lightweight version of the architecture that retains the core principles of our approach while significantly reducing the computational overhead. This would broaden the applicability of our method to a wider range of practical scenarios where both high accuracy and rapid inference are required.

V. Conclusion

In this article, aiming at the problems of easily missing low-difference changes and easily misdetecting high-difference nonchanges in RSSCD, we propose a spatiotemporal multidimensional scanning method based on Mamba. First, this method extracts the information of bitemporal images in the temporal dimension through frame interpolation. Then, with the help of Mamba, on the one hand, it strengthens the feature expression ability through global context modeling, and on the other hand, it establishes a progressive change perception mechanism to accurately capture subtle temporal evolution information. Next, a method of MSOS is proposed. Specifically, by improving the patch embedding module, features with seven additional receptive fields are added to the binary change detection branch and the semantic segmentation branch, respectively. At the same time, we also introduce more scanning directions in the Mamba-based calculation to construct an omnidirectional MS feature perception network. Finally, aiming at the problem of class imbalance in RSSCD, we propose CDW Loss to achieve breakthroughs from two dimensions: feature constraints and weight optimization. We have verified the effectiveness of our method on three datasets with different resolutions. The proposed method achieves significant improvements in the SeK metric, a comprehensive indicator for semantic change detection, outperforming the current state-of-the-art methods by 17.67%, 4.54%, and 20.80% on the Landsat–SCD, SECOND, and Hi-UCD datasets, respectively, which demonstrates the effectiveness of our method on data of various resolutions, particularly low-, medium-, and ultrahigh-resolutions by extracting and utilizing richer temporal dimension information.

References

[1] R. Caye Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

[2] A. Graves, "Generating sequences with recurrent neural networks," 2013, *arXiv:1308.0850*.

[3] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[4] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.

[5] X. Shi, Z. Chen, H. Wang, D. Yeung, W. K. Wong, and W. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 802–810.

[6] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, and P. He, "SCDNET: A novel convolutional network for semantic change detection in high resolution optical remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102465.

[7] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[8] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[9] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[10] L. Huang, B. Jiang, S. Lv, Y. Liu, and Y. Fu, "Deep-learning-based semantic segmentation of remote sensing images: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8370–8396, 2024.

[11] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[12] K. Yue, L. Yang, R. Li, W. Hu, F. Zhang, and W. Li, "TreeUNet: Adaptive tree convolutional neural networks for subdecimeter aerial image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 156, pp. 1–13, Oct. 2019.

[13] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[15] X. Sun, M. Xia, and T. Dai, "Controllable fused semantic segmentation with adaptive edge loss for remote sensing parsing," *Remote Sens.*, vol. 14, no. 1, p. 207, Jan. 2022.

[16] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[17] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, May 2021.

[18] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[19] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2020, pp. 4095–4104.

[20] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[21] R. Xu et al., "RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 1052–1064, 2023, doi: 10.1109/TIP.2023.3238648.

[22] J. Lv, Q. Shen, M. Lv, Y. Li, L. Shi, and P. Zhang, "Deep learning-based semantic segmentation of remote sensing images: A review," *Frontiers Ecology Evol.*, vol. 11, Jul. 2023, Art. no. 1201125.

[23] Z. Zhong et al., "Understanding imbalanced semantic segmentation through neural collapse," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2023, pp. 19550–19559.

[24] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[25] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4353–4361.

[26] Q. Wang, X. Zhang, G. Chen, F. Dai, Y. Gong, and K. Zhu, "Change detection based on faster R-CNN for high-resolution remote sensing images," *Remote Sens. Lett.*, vol. 9, no. 10, pp. 923–932, Oct. 2018.

[27] P. Han, C. Ma, Q. Li, P. Leng, S. Bu, and K. Li, "Aerial image change detection using dual regions of interest networks," *Neurocomputing*, vol. 349, pp. 190–201, Sep. 2019.

[28] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.

[29] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[30] B. Wijenayake et al., "Mamba-FCS: Joint spatio- frequency feature fusion, change-guided attention, and SeK loss for enhanced semantic change detection in remote sensing," 2025, *arXiv:2508.08232*.

[31] T. Wang et al., "AtrousMamaba: An Atrous-window scanning visual state space model for remote sensing change detection," 2025, *arXiv:2507.16172*.

[32] Z. Xu, H. Wu, W. Jiang, and J. Geng, "Graph aggregation prototype learning for semantic change detection in remote sensing," 2025, *arXiv:2507.10938*.

[33] S. Tian, X. Tan, A. Ma, Z. Zheng, L. Zhang, and Y. Zhong, "Temporal-agnostic change region proposal for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 306–320, Oct. 2023.

[34] A. Gu, T. Dao, S. Ermon, A. Rudra, and C. Ré, "HiPPO: Recurrent memory with optimal polynomial projections," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 1474–1487.

[35] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "FlashAttention: Fast and memory-efficient exact attention with IO-awareness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 16344–16359.

[36] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 62429–62442.

[37] G. Huang, Y. Sun, Z. Liu, D. Sedra, and K. Q. Weinberger, "Deep networks with stochastic depth," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 646–661.

[38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[39] M. Lin, G. Yang, and H. Zhang, "Transition is a process: Pair-to-video change detection networks for very high resolution remote sensing images," *IEEE Trans. Image Process.*, vol. 32, pp. 57–71, 2023.

[40] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[41] Y. Lee, J. Kim, J. Willette, and S. J. Hwang, "MPViT: Multi-path vision transformer for dense prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2022, pp. 7277–7286.

[42] M. Berman, A. R. Triki, and M. B. Blaschko, "The Lovasz-softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4413–4421.

[43] P. Yuan, Q. Zhao, X. Zhao, X. Wang, X. Long, and Y. Zheng, "A transformer-based Siamese network and an open optical dataset for semantic change detection of remote sensing images," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 1506–1525, Dec. 2022.

[44] S. Tian, Y. Zhong, Z. Zheng, A. Ma, X. Tan, and L. Zhang, "Large-scale deep learning based binary and semantic change detection in ultra high resolution remote sensing imagery: From benchmark datasets to urban application," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 164–186, Nov. 2022.

[45] L. Ding, J. Zhang, H. Guo, K. Zhang, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610814.

[46] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

[47] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[48] K. Tang, F. Xu, X. Chen, Q. Dong, Y. Yuan, and J. Chen, "The ClearSCD model: Comprehensively leveraging semantics and change relationships for semantic change detection in high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 299–317, May 2024.

[49] Z. Li, X. Wang, S. Fang, J. Zhao, S. Yang, and W. Li, "A decoder-focused multitask network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610907.

[50] J. Long, M. Li, X. Wang, and A. Stein, "Semantic change detection using a hierarchical semantic graph interaction network from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 318–335, May 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271624001709

[51] S. Fang, W. Li, S. Yang, Z. Li, J. Zhao, and X. Wang, "BT-HRSCD: high-resolution feature is what you need for a semantic change detection network with a triple-decoding branch," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611203.

[52] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:35:43 UTC from IEEE Xplore. Restrictions apply.